WHITE PAPER · HWACHEON HI-ECO 21 HS × AEROSPACE

Hwacheon Hi-Eco 21 HS for Aerospace

High-speed CNC lathe for small-part precision turning where accuracy and cycle time matter most. This paper covers what the machine does, the aerospace parts we produce with it, the alloys we run, the tolerances and finishes aerospace buyers expect, and the documentation package.

8" OD × 12" L max turning ENVELOPE	**±0.0005" routine on critical features** TOLERANCE	**1994** SINCE
Same-week / rig-down direct RESPONSE		



Bearkat Blur · 8" OD × 12" L max turning · high-speed precision CNC turning center

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Executive summary

Machine: Hwacheon Hi-Eco 21 HS (shop nickname: *Bearkat Blur*). high-speed precision CNC turning center at B&R Productions, running from our New Waverly, Texas shop.

Application: High-speed CNC lathe for small-part precision turning where accuracy and cycle time matter most for Aerospace customers — Aerospace primes, tier-1 and tier-2 suppliers, MRO shops, propulsion component manufacturers, structural assembly integrators, actuator OEMs.

Envelope: 8" OD × 12" L max turning. **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

The machine, in context

Bearkat Blur exists because certain oilfield and aerospace parts are small, high-quantity, and can't tolerate the setup time of a larger machine. Bushings, couplings, precision shafts, valve inserts, and any repeatable geometry in the 8-inch envelope — this machine cycles fast without giving up accuracy. When cycle time drives the quote, this is the machine that wins.

Why Bearkat Blur for Aerospace work

Small aerospace turned parts — bushings, spacers, precision inserts, small shafts — are ordered in quantity with tight tolerances and no tolerance at all for batch drift. An 8" × 12" high-speed cell holding ±0.0005" while cycling fast is the right economics for that work.

What this looks like on a real part

A batch of titanium bushings is a heat management problem at small scale. Titanium wants aggressive chip load and flood coolant — light cuts overheat it and cook the tool, which is the opposite of most operators' instinct on a small precision part. Get the feed too timid and you glaze the surface, work-harden the next pass, and watch the tool die early. The intuition that works on steel actively hurts here.

> **When this is the wrong machine**
>
> 8 inches is a hard wall, and 15 HP continuous is finish-oriented. Larger aerospace turned parts go to a Hi-Eco 35 cell; anything requiring milled features in the same setup goes to Torque & Tumble.

Full specifications

Model	Hwacheon Hi-Eco 21 HS ("Bearkat Blur")

Type	High-speed CNC turning center — small-part specialist
Max turning capacity	8" OD × 12" L (B&R working spec)
Platform max swing	22.8" over bed; 11.02" turning diameter platform-max
Distance between centers	15.74" platform
Bar capacity	2" (through-spindle)
Chuck size	8"
Spindle nose	A2-6; 2.4" spindle bore
Spindle speed	4,000 RPM max
Spindle power	15 HP continuous / 20 HP 30-min rating
Turret	12-station
Rapid traverse	1,181 IPM on both X and Z axes
Control	Fanuc 18-T
Machine weight	~8,900 lb
Tolerance capability	±0.0005" routine on critical features
Materials	Stainless, brass, tool steel, Inconel 718/625, 17-4 PH, Monel

Values marked "platform" reflect the machine manufacturer's published spec range for this platform family; B&R's working spec is what we quote on this specific unit.

Who this serves in Aerospace

Typical buyer: Aerospace primes, tier-1 and tier-2 suppliers, MRO shops, propulsion component manufacturers, structural assembly integrators, actuator OEMs.

What's hard about aerospace work: Certification-driven documentation, exotic superalloys, tight tolerances on structural and engine components, and finding a job shop that understands aged-condition superalloys rather than learning on the part.

Typical Aerospace parts we produce on Bearkat Blur

- Precision bushings and pins
- Small aerospace shafts and precision round parts
- Actuator small-part components
- Fittings and adapters for fluid and pneumatic systems
- Bearing components and sleeves

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

Alloys we run for Aerospace

Standard range: Titanium Grade 5 (Ti-6Al-4V) and Grade 23 (Ti-6Al-4V ELI) · Inconel 718 · Inconel 625 · Waspaloy · A286 · 17-4 PH · 15-5 PH · 13-8 Mo PH · Aluminum 7075/6061 · Custom aerospace alloys on request.

Titanium Grade 5 is the workhorse and it's temperature-sensitive: aggressive chip load with flood coolant keeps the heat in the chip, not the workpiece. Light cuts overheat titanium and cook tools; that's why titanium demands a different intuition than steel. Inconel 718 in aerospace-aged condition is harder and less forgiving than the annealed billet stock — most aerospace 718 work assumes aged. 17-4 PH and 15-5 PH are common for structural and actuator components; both are precipitation-hardening stainless with condition-specific machining plans.

> **What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

How we machine it — our 5-step process

This is the process B&R runs on every job, aerospace or otherwise. The specifics of feeds, speeds, and tooling shift by alloy and machine; the discipline is universal.

1. **Material verification.** Every job starts with material traceability. Heat and lot numbers documented against the mill test report. For customer-supplied material, we verify against the CofC before touching the stock.

2. **First-article layout.** Stock is measured and laid out before any cutting starts. Concentricity, roundness, and squareness of the starting material established; if the stock is out of tolerance we call the customer before wasting time on a bad blank.

3. **Roughing with process control for the alloy.** Feeds, speeds, and coolant strategy set for the specific alloy family — sharp coated carbide (or PCD for high-volume Ti Grade 5), positive rake, high-pressure through-tool coolant on 2507 and Inconel to control cutting temperature. Interrupted cuts get extra attention because they cycle heat into the workpiece.

4. **Finish pass to spec.** Finish pass with a fresh insert, controlled DOC, and measured surface-finish targets. Critical features (bore concentricity, seat grooves, sealing surfaces) are held tighter than most job-shop lathes can offer.

5. **CMM verification + documentation.** Every critical dimension CMM-verified. First-article report generated. Certificate of conformance issued with delivery. If your quality group needs a documentation package beyond that, tell us with the RFQ and we'll be straight about what we can and can't produce in-house.

Tolerance and finish expectations in Aerospace

±0.0005" routine on critical features. Tighter with proper fixture and setup — some aerospace features run ±0.0002" or GD&T-driven cylindricity and position tolerances. CMM verification standard on first articles; process-driven measurement on runs. Surface finish measured, not estimated.

Documentation and quality workflow

First-article layout and CMM verification standard. Material traceability by heat and lot. Certificates of conformance issued with delivery. Proprietary prints handled under NDA — prints stay in-shop and aren't sent to outside vendors. To be clear about what we are not: B&R is not AS9100-registered and not ITAR-registered. If

your program requires either, you need a registered shop and we'll say so on the first call rather than after you've placed the order.

Response time and emergency work

AOG (aircraft-on-ground) work prioritized when material is available. Emergency response supported for existing aerospace customers with blanket arrangements or established relationships. Call direct at (936) 291-7827.

How we compare

Compared to a captive aerospace shop: we're faster for one-offs and small runs, more flexible on setup, and we don't require a long approval process. Compared to a lowest-bidder commodity shop: we understand the alloys and we treat first-article inspection as a discipline rather than a formality. Where a registered shop beats us is on programs that require AS9100 flow-down — we're the wrong call for those, and the right call for non-controlled work, prototypes, tooling, and MRO parts where alloy competence and turnaround matter more than a certificate on the wall.

Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request, and prints stay in-shop rather than going out to third-party vendors. Note for defense and aerospace buyers: B&R is not ITAR- or AS9100-registered, so export-controlled prints and AS9100 flow-down programs need a registered shop — we'll tell you that on the first call, not after the PO.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Frequently asked questions

8 questions covering the Hwacheon Hi-Eco 21 HS platform and Aerospace work. Also indexed as FAQ schema for AI answer engines.

What's the max turning diameter on the Hi-Eco 21 HS?

8" OD × 12" L is B&R's working spec. The platform itself has a 22.8" swing over bed and can accept turning diameter up to 11.02" with 15.74" between centers. Bar capacity: 2" through spindle.

How fast is this machine — what's the spindle RPM?

4,000 RPM max spindle speed. Rapid traverse 1,181 IPM on both X and Z. Combined with the 8" chuck this is our high-cycle-count small-part specialist.

What's the turret capacity?

12-station turret. Enough tool positions for most single-setup precision turning jobs including grooving, threading, and finish operations.

Do you serve aerospace primes and tier-1/2 suppliers?

On work that doesn't require a registered QMS, yes — that's the honest boundary. B&R is not AS9100-registered, and where that matters it matters absolutely, so we say it before you spend time on an RFQ: if your program requires AS9100 flow-down, we're the wrong shop. Prototypes, tooling, fixtures, MRO and repair parts, and non-controlled components are where a shop like this is genuinely useful to a prime or a tier-1, and that's the work we'd like to quote.

Can you handle ITAR-controlled aerospace/defense work?

No. ITAR compliance requires registration with the U.S. State Department's DDTC, and B&R does not hold that registration. If your print is ITAR-controlled, you need a registered shop — send it elsewhere and we'd rather tell you now than after you've placed an order. Plenty of aerospace and defense work isn't export-controlled, and that work we quote gladly. Either way, proprietary prints are handled under NDA and stay in-shop.

What tolerances do you hold on aerospace parts?

±0.0005" routine on critical features. First-article layout and CMM verification standard. Surface-finish targets measured to spec.

What aerospace alloys do you run?

Titanium Grade 5 (Ti-6Al-4V), Grade 23 ELI, Inconel 718 and 625, Waspaloy, A286, 17-4 PH, 15-5 PH, aluminum 7075/6061. Custom aerospace alloys on request.

Can you respond to AOG (aircraft-on-ground) work?

Yes, when material is available. Emergency response supported for existing aerospace customers. Newer customers should call to discuss what's realistic for their specific alloy and part.

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Ready to talk about your Aerospace work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

Published by B&R Productions — a precision CNC machining shop in New Waverly, Texas, in business since 1994. Serving oil & gas, aerospace, defense, and industrial customers across Texas and the Gulf Coast.

Written by the B&R Productions team. Machine specs verified against manufacturer data. Alloy and process notes reflect shop practice as of 2026-01-15.